P0 Walk-Forward Validation

Point-in-time backtest of price-derived factor blocks | 2020-07-01 to 2025-01-31 | monthly rebalance, 55 dates | generated 2026-08-20T17:02:10

What this report does and does not establish

This is not a validation of Model 5.0. It exercises only the price-derived blocks. Quality, growth and value together carry 70% of the production score and require point-in-time fundamentals that are not yet ingested, so none of them are tested here.

It answers one question from p0.md §7E: **how much of the model's behaviour is momentum?** A momentum-only ranking that performs close to the full model would mean the fundamental blocks are decoration. Read every number below as evidence about momentum and risk, not about Model 5.0.

Two further gaps: these ablations rank **market-wide** where production ranks within sector, and momentum-only omits RS_Sector_6M_Pct (20% of the production momentum block) because the bhavcopy archive carries no sector classification.

Integrity controls

Control	Status
Universe source	Archived NSE bhavcopy per session (point-in-time, includes delisted names)
Survivorship	3384 securities; 522 delisted retained in-sample
Identifier	ISIN-bridged Security_ID; 419 face-value changes linked
Execution	Signal on close of t, fill at open of next confirmed session
Corporate actions	659 split/bonus adjusted, 11580 dividends, 620 unadjustable events excluded
Delisting policy	haircut@0.50
Costs	effective-dated Indian charges + 10bp half-spread + sqrt impact at Rs 1.00 lakh/position
Eligibility	min turnover Rs 20.00 lakh/day, trading frequency >= 0.8, history >= 200 sessions, INE-class only (ETFs excluded)
Parameters	Frozen; no re-optimisation across the window

Eligible universe per rebalance

Metric	Value
Rebalance dates	55
Mean eligible	1346
Min / max eligible	892 / 1682
Mean considered before filters	1817

Rank information coefficient — Gross of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	55	0.0049	0.0086	58%	-0.1450	0.1777
growth only	3M	55	0.0062	0.0047	53%	-0.1347	0.1112
growth only	6M	55	0.0011	-0.0003	49%	-0.1043	0.1207
model 5	1M	55	0.0324	0.0389	65%	-0.3275	0.2845
model 5	3M	55	0.0614	0.0632	78%	-0.2115	0.2996
model 5	6M	55	0.0774	0.0744	87%	-0.2914	0.2626
model 5 gated	1M	55	0.0301	0.0373	65%	-0.3275	0.2845
model 5 gated	3M	55	0.0586	0.0574	76%	-0.2115	0.2996
model 5 gated	6M	55	0.0747	0.0744	87%	-0.2914	0.2460
momentum only	1M	55	0.0234	0.0231	60%	-0.3289	0.2515
momentum only	3M	55	0.0547	0.0544	75%	-0.1706	0.2399
momentum only	6M	55	0.0747	0.0982	80%	-0.3062	0.2626
quality only	1M	55	0.0158	0.0097	51%	-0.2126	0.2553
quality only	3M	55	0.0100	0.0221	58%	-0.2169	0.2387
quality only	6M	55	0.0013	-0.0037	49%	-0.2188	0.1837
random ranking	1M	55	-0.0009	0.0014	51%	-0.0642	0.0743
random ranking	3M	55	0.0008	-0.0009	47%	-0.0668	0.0642
random ranking	6M	55	-0.0006	-0.0001	49%	-0.0664	0.0515
simple quality momentum	1M	55	0.0221	0.0203	56%	-0.3389	0.2737
simple quality momentum	3M	55	0.0539	0.0454	71%	-0.1775	0.2307
simple quality momentum	6M	55	0.0751	0.0806	84%	-0.3043	0.2220
value only	1M	55	0.0345	0.0347	73%	-0.0889	0.1463
value only	3M	55	0.0515	0.0457	85%	-0.1001	0.1871
value only	6M	55	0.0635	0.0572	89%	-0.0818	0.1956

Portfolio and bucket evidence — Gross of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	3.02%	3.02%	+0.00%	-0.212	+0.07%
equal weight universe	3M	8.59%	8.59%	+0.00%	-0.103	+0.28%

equal weight universe	6M	18.13%	18.13%	+0.00%	-0.103	+1.10%
growth only	1M	3.02%	3.60%	+0.58%	0.030	-0.04%
growth only	3M	8.59%	9.51%	+0.92%	0.091	-0.21%
growth only	6M	18.13%	18.51%	+0.38%	-0.103	-1.54%
model 5	1M	3.02%	4.46%	+1.44%	0.867	+0.80%
model 5	3M	8.59%	12.78%	+4.19%	0.976	+3.33%
model 5	6M	18.13%	23.69%	+5.57%	0.988	+6.41%
model 5 gated	1M	3.02%	4.19%	+1.18%	0.939	+0.69%
model 5 gated	3M	8.59%	11.89%	+3.30%	0.927	+2.96%
model 5 gated	6M	18.13%	22.58%	+4.45%	0.927	+5.96%
momentum only	1M	3.02%	4.15%	+1.14%	0.891	+1.58%
momentum only	3M	8.59%	11.34%	+2.75%	0.976	+6.62%
momentum only	6M	18.13%	21.37%	+3.24%	1.000	+11.40%
quality only	1M	3.02%	2.52%	-0.50%	-0.867	-0.73%
quality only	3M	8.59%	7.68%	-0.91%	-0.867	-2.22%
quality only	6M	18.13%	17.11%	-1.02%	-0.939	-5.64%
random ranking	1M	3.02%	2.87%	-0.15%	-0.382	+0.06%
random ranking	3M	8.59%	8.84%	+0.25%	-0.176	+0.42%
random ranking	6M	18.13%	18.16%	+0.03%	0.176	+0.18%
simple quality momentum	1M	3.02%	3.06%	+0.04%	0.952	+0.99%
simple quality momentum	3M	8.59%	9.40%	+0.81%	0.988	+4.46%
simple quality momentum	6M	18.13%	17.67%	-0.46%	0.964	+8.08%
value only	1M	3.02%	5.69%	+2.67%	0.527	+0.79%
value only	3M	8.59%	17.45%	+8.86%	0.576	+2.68%
value only	6M	18.13%	38.85%	+20.72%	0.624	+5.23%

Rank information coefficient — Net of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	55	0.0162	0.0208	62%	-0.1307	0.1831
growth only	3M	55	0.0129	0.0161	60%	-0.1286	0.1168
growth only	6M	55	0.0056	0.0034	53%	-0.0999	0.1285
model 5	1M	55	0.0553	0.0559	75%	-0.3113	0.2952
model 5	3M	55	0.0747	0.0789	82%	-0.2046	0.3078

model 5	6M	55	0.0864	0.0859	89%	-0.2843	0.2770
model 5 gated	1M	55	0.0530	0.0559	75%	-0.3113	0.2948
model 5 gated	3M	55	0.0720	0.0762	82%	-0.2046	0.3078
model 5 gated	6M	55	0.0837	0.0859	89%	-0.2843	0.2604
momentum only	1M	55	0.0372	0.0361	64%	-0.3175	0.2550
momentum only	3M	55	0.0628	0.0697	76%	-0.1673	0.2465
momentum only	6M	55	0.0801	0.1014	82%	-0.3009	0.2703
quality only	1M	55	0.0407	0.0356	67%	-0.1946	0.2697
quality only	3M	55	0.0241	0.0353	65%	-0.2049	0.2489
quality only	6M	55	0.0109	0.0052	51%	-0.2107	0.1976
random ranking	1M	55	-0.0008	0.0005	53%	-0.0624	0.0750
random ranking	3M	55	0.0010	-0.0009	47%	-0.0688	0.0624
random ranking	6M	55	-0.0007	-0.0006	49%	-0.0678	0.0519
simple quality momentum	1M	55	0.0405	0.0408	65%	-0.3247	0.2851
simple quality momentum	3M	55	0.0647	0.0629	76%	-0.1732	0.2375
simple quality momentum	6M	55	0.0823	0.0876	85%	-0.2980	0.2348
value only	1M	55	0.0324	0.0368	73%	-0.0834	0.1378
value only	3M	55	0.0502	0.0424	85%	-0.0955	0.1843
value only	6M	55	0.0627	0.0562	89%	-0.0778	0.1927

Portfolio and bucket evidence — Net of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	1.22%	1.22%	+0.00%	-0.151	+0.12%
equal weight universe	3M	6.79%	6.79%	+0.00%	-0.103	+0.33%
equal weight universe	6M	16.33%	16.33%	+0.00%	-0.103	+1.15%
growth only	1M	1.22%	2.02%	+0.80%	0.636	+0.34%
growth only	3M	6.79%	7.93%	+1.14%	0.345	+0.17%
growth only	6M	16.33%	16.91%	+0.58%	-0.030	-1.16%
model 5	1M	1.22%	3.05%	+1.83%	0.988	+1.66%
model 5	3M	6.79%	11.36%	+4.57%	0.976	+4.18%
model 5	6M	16.33%	22.27%	+5.94%	0.988	+7.26%
model 5 gated	1M	1.22%	2.82%	+1.60%	1.000	+1.57%
model 5 gated	3M	6.79%	10.51%	+3.72%	0.927	+3.84%
model 5 gated	6M	16.33%	21.21%	+4.87%	0.927	+6.84%
momentum only	1M	1.22%	2.38%	+1.16%	0.952	+1.93%
momentum only	3M	6.79%	9.57%	+2.78%	0.976	+6.97%
momentum only	6M	16.33%	19.60%	+3.27%	1.000	+11.75%

quality only	1M	1.22%	1.30%	+0.08%	0.030	+0.20%
quality only	3M	6.79%	6.45%	-0.34%	-0.709	-1.29%
quality only	6M	16.33%	15.88%	-0.45%	-0.782	-4.71%
random ranking	1M	1.22%	1.06%	-0.16%	-0.333	+0.04%
random ranking	3M	6.79%	7.05%	+0.25%	-0.236	+0.41%
random ranking	6M	16.33%	16.36%	+0.03%	0.176	+0.17%
simple quality momentum	1M	1.22%	1.63%	+0.41%	0.964	+1.59%
simple quality momentum	3M	6.79%	7.97%	+1.17%	0.988	+5.06%
simple quality momentum	6M	16.33%	16.23%	-0.11%	0.964	+8.68%
value only	1M	1.22%	3.76%	+2.54%	0.624	+1.20%
value only	3M	6.79%	15.52%	+8.73%	0.527	+3.09%
value only	6M	16.33%	36.93%	+20.60%	0.624	+5.64%

Turnover

One-way turnover of a top-20 portfolio per rebalance. 1.0 means the portfolio was fully replaced. High turnover is what converts a real gross edge into no net edge.

Strategy	Mean one-way turnover	Rebalances
equal weight universe	0.060	55
growth only	0.118	55
model 5	0.332	55
model 5 gated	0.388	55
momentum only	0.486	55
quality only	0.082	55
random ranking	0.970	55
simple quality momentum	0.391	55
value only	0.164	55

Fill coverage

Share of scored positions that produced a usable forward return. A dropped position is recorded with a reason rather than silently omitted; a thin horizon here weakens every statistic above it.

Horizon	Coverage	Positions	Main reasons dropped
1M	99.6%	633204	unadjustable_action 2268, no_entry_price 54
3M	98.9%	633204	unadjustable_action 6633, no_entry_price 54
6M	97.9%	633204	unadjustable_action 13077, no_entry_price 54

How to read this

The random_ranking row is the calibration. It is a seeded random score, so its IC is what "no signal" measures on this universe and horizon. Any strategy whose IC is not clearly separated from it has not demonstrated anything.

equal_weight_universe scores every name identically, so its rank IC is undefined rather than zero. It is present as the return benchmark from p0.md §7B, not as a ranking.

Judge consistency before magnitude. A mean IC of 0.03 with 70% of periods positive is stronger evidence than a mean of 0.06 driven by two periods. Check the %+ and worst columns before the mean.

Net is the number that matters. If a gross edge disappears net of costs, the ranking may still be real but is not tradable at this position size and turnover.

Known limitations of this run

Restatement bias remains. No fundamentals are used here, so it does not bite yet — but it will once the XBRL panel lands, for any period falling back to current values rather than as-filed ones.

Delisting recovery is an assumption, not an observation. The bhavcopy does not record why a security stopped trading. Re-run with --delisting-strategy zero and last_close to bound how much the headline depends on it.

Period count is modest. Monthly rebalancing over roughly four years is enough to judge cross-sectional rank IC and its stability. It is not enough to claim regime robustness across a full cycle.

Overlapping horizons are not independent. A 6M horizon sampled monthly reuses most of its window, so any t-statistic on per-period ICs overstates significance and is reported as a rough guide only.

Cost assumptions are documented estimates. Brokerage defaults to zero (discount-broker delivery); half-spread and impact coefficient are modelled, not measured. Gross and net are shown separately for that reason.

Next step

Phase B: ingest NSE XBRL filings into a point-in-time fundamental panel keyed on filing timestamp, which unlocks the quality, growth and value blocks and makes a genuine Model 5.0 walk-forward possible. Until then the fundamental 70% of the score remains untested.